



Secure and Scalable Healthcare Appointment Scheduling Through Distributed Backend Development

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Abstract

CarePlus is a web-based healthcare appointment scheduling system that allows patients to register, log in, search for doctors, view availability, and book appointments. The system uses a React frontend, FastAPI backend, and MySQL database with JWT-based authentication and role-based access. The system simplifies appointment management while providing administrators with centralized control over doctors, appointments, holidays, notifications, and payments.

Problem Statement & Objectives

Problem

Traditional appointment scheduling can require manual coordination and makes it difficult to manage doctor availability, appointment slots, patient records, and booking conflicts efficiently.

Objectives

- Provide an easy online appointment booking system.
- Manage doctor availability and appointment slots.
- Implement secure patient and administrator authentication.
- Provide centralized appointment, notification, and payment management

Existing vs Proposed System

Feature	Existing System	Proposed System
Appointment Booking	Manual / limited online coordination	Online slot-based booking
Doctor Availability	Difficult to track	Calendar and availability management
Authentication	Basic/manual access	JWT-based authentication
Appointment Management	Separate/manual records	Centralized database
Payment	Manual processing	Demo payment + receipt

Feasibility Study

Technical

Uses commonly available technologies such as React.js, FastAPI, Python, MySQL, REST APIs, JWT, and SQLAlchemy.

Economic

Uses an open-source technology stack, reducing software and development costs.

Operational

Provides separate patient and administrator workflows with a simple web interface.

Schedule

The system can be developed and tested within the academic semester timeline.

Results & Testing

- Patient login and appointment booking work successfully.
- Doctor availability and booking validation are implemented.
- Notifications and demo payments are included.
- Admin can manage doctors, appointments, and payments.

Conclusion

CarePlus provides a simple and centralized healthcare appointment scheduling system. It enables patients to book appointments and helps administrators manage doctors, availability, appointments, notifications, and payments efficiently. The system integrates the React frontend, FastAPI backend, and MySQL database.

Tools & Technologies

Frontend

React, JavaScript, HTML5, CSS3, Vite

Backend

Python, FastAPI, Uvicorn, REST APIs

Database

MySQL, SQLAlchemy, PyMySQL

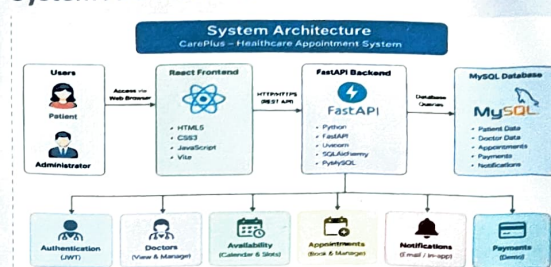
Security

JWT, Passlib, bcrypt, Bearer Authentication

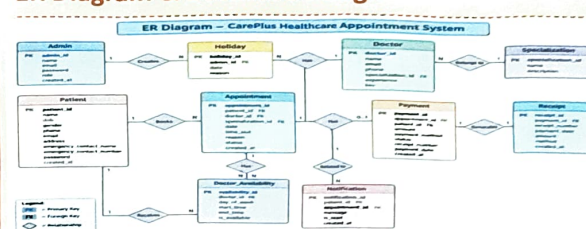
Development Tools

VS Code, Git/GitHub, MySQL

System Architecture



ER Diagram & Database Design



Implementation Screenshots



Future Scope

- Integrate a real payment gateway.
- Add a separate doctor login and portal.
- Enable cloud deployment and scalability.
- Add advanced reminders and notifications.

Handwritten signatures and notes: 'vgood' and 'H. Sharan'.